CRIMA ontology discussion: a focus on IC

19th January 2026

UNIBZ, LOA CNR-ISTC

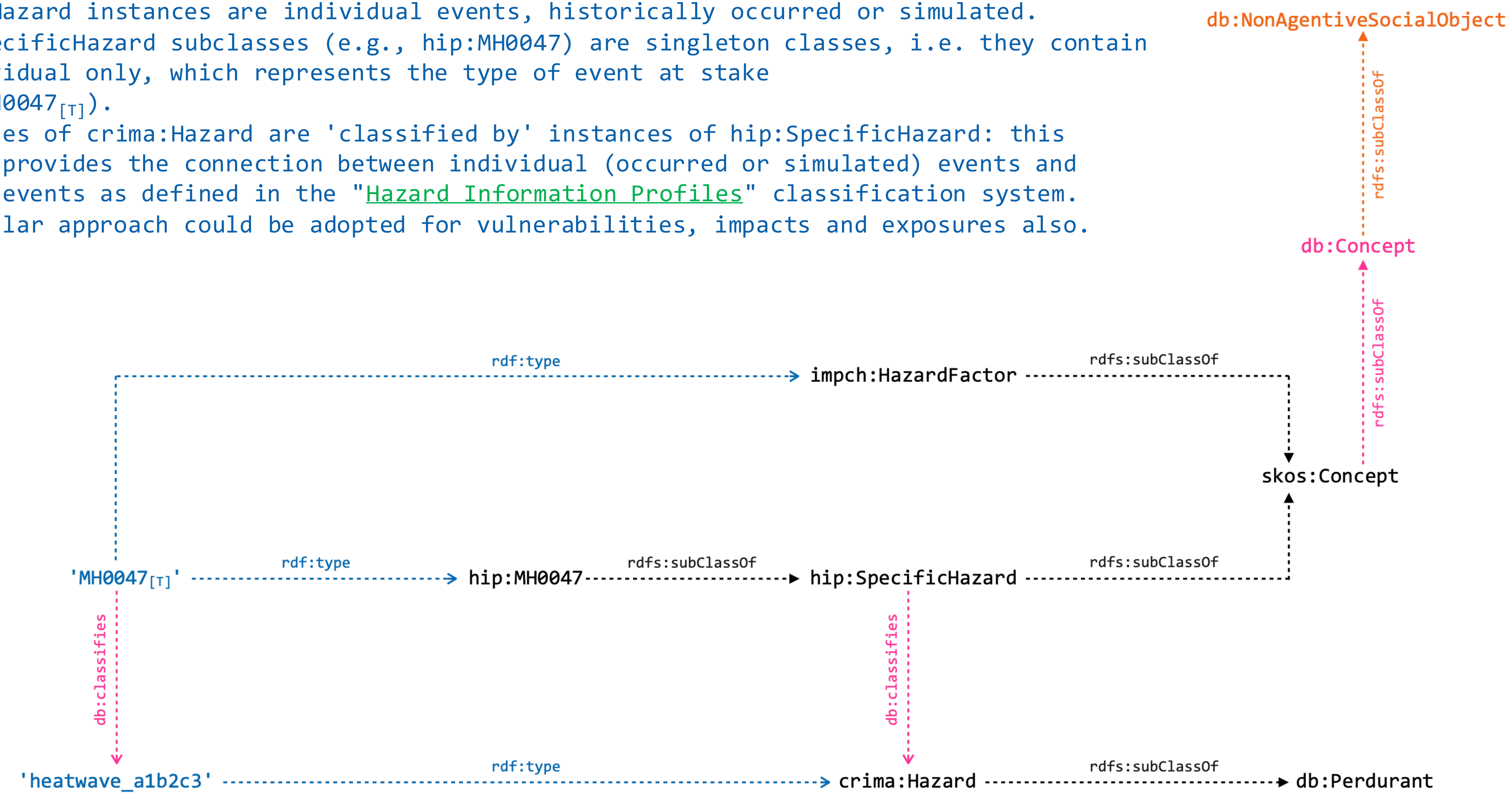
Outline

- **Theoretical and factual knowledge**: Types, individuals and types-as-individuals.
- **Chains of events and processes**: Potential impacts and their cause-and-effect relationships.
- Perdurants and their **participants**.
- **Domain entities and their properties**: The SOSA 'observation' pattern.
- **Vulnerabilities** as endurant properties: Intrinsic, Extrinsic, proxies and triggering factors.
- Being vulnerable and exposed as derived properties: **A reasoning example**.
- **Impact Chain connections** and their semantics.
- **Impact classification**: The IPCC WGII AR6 proposal.
- Impact Chain **labelling conventions**.

Theoretical knowledge:

Types, individuals and types-as-individuals

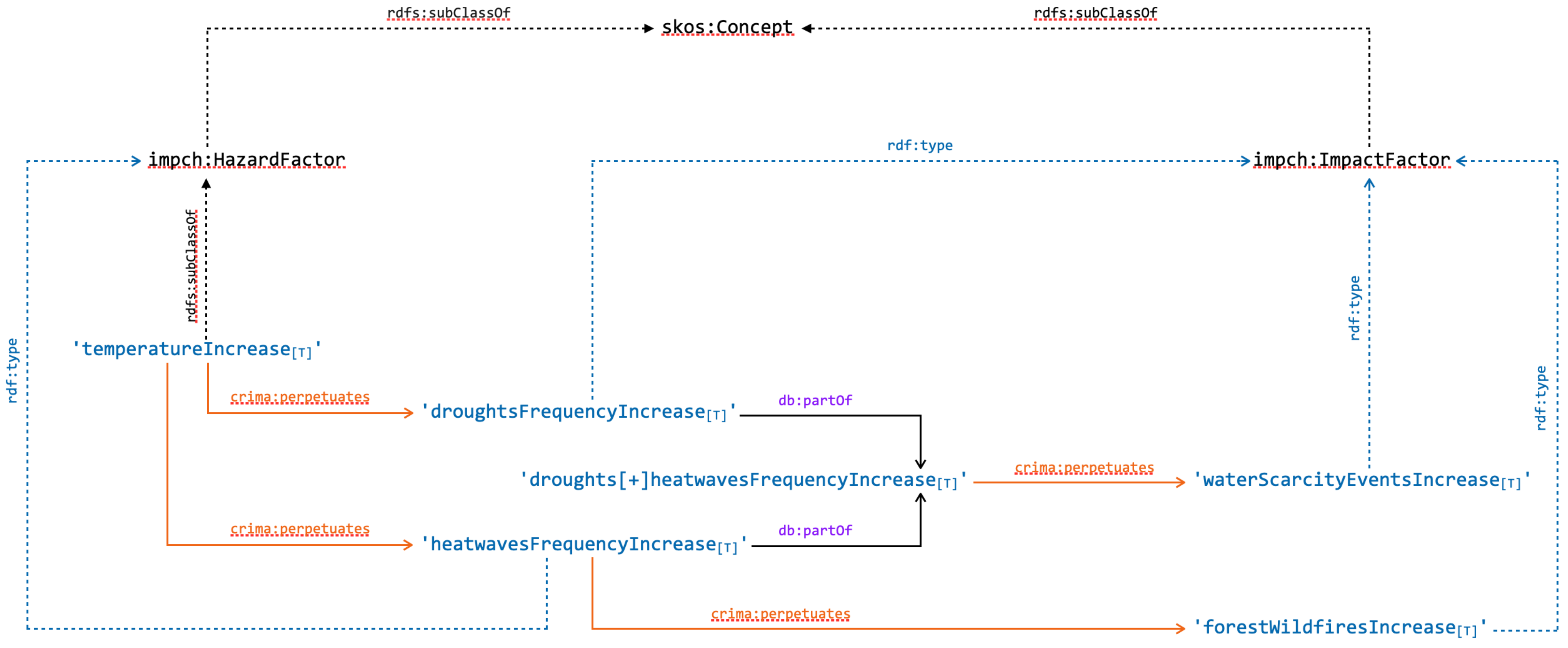
• Impact chain nodes represent types (of events, exposures, vulnerabilities, etc.).
• Individuals (or, instances) are in blue. Individuals with suffix '[T]' are meant to represent types-as-individuals.
• crima:Hazard instances are individual events, historically occurred or simulated.
• hip:SpecificHazard subclasses (e.g., hip:MH0047) are singleton classes, i.e. they contain one individual only, which represents the type of event at stake (e.g., $\text{MH0047}_{[T]}$).
• Instances of crima:Hazard are 'classified by' instances of hip:SpecificHazard: this relation provides the connection between individual (occurred or simulated) events and types of events as defined in the "Hazard Information Profiles" classification system.
-> A similar approach could be adopted for vulnerabilities, impacts and exposures also.



Chains of perdurants:

Potential impacts and their cause-and-effect relationships

- Process chain involving hazards and impacts.
- Impact chain examples show that perdurants (see, here, impacts) can combine to give rise to more complex perdurants (whose nature cannot be simply reduced to the sum of their constituents).
- The semantics of the connections here introduced is discussed in the following slides.
- (Notice that 'temperature increase' is not among the hazards listed in the UNDRR-HIP).



Perdurants and their participants

3.5. Participation

The participation (PC) relation connects endurants, perdurants, and times, i.e. endurants *participate* in perdurants at a certain time (a6). Here we write $\mathsf{PC}(x, y, t)$ for 'x participates in y at time t'. (a7) states that a perdurant has at least one participant and (a8) that an endurant participates in at least one perdurant. Axiom (a9) says that for an endurant to participate in a perdurant they must be present at the same time. We also introduce the relation of constant participation ($\mathsf{PC_C}$), cf. (d10), i.e., participation during the whole perdurant, which we will use in Sections 4.4 and 4.5.

a6	$\mathsf{PC}(x, y, t) \rightarrow \mathrm{ED}(x) \wedge \mathrm{PD}(y) \wedge \mathrm{T}(t)$	(*Participation typing*, cf. Ad33)
a7	$\mathrm{PD}(x) \wedge \mathsf{PRE}(x, t) \rightarrow \exists y(\mathsf{PC}(y, x, t))$	(cf. Ad34)
a8	$\mathrm{ED}(x) \rightarrow \exists y, t(\mathsf{PC}(x, y, t))$	(cf. Ad35)
a9	$\mathsf{PC}(x, y, t) \rightarrow \mathsf{PRE}(x, t) \wedge \mathsf{PRE}(y, t)$	(cf. Ad36)
a10	$\mathsf{PC_C}(x, y) \stackrel{\text{def}}{=} \exists t(\mathsf{PRE}(y, t)) \wedge \forall t(\mathsf{PRE}(y, t) \rightarrow \mathsf{PC}(x, y, t))$	(*Const. Participation*, cf. Dd63)

Applied Ontology 17 (2022) 45–69
DOI 10.3233/AO-210259
IOS Press

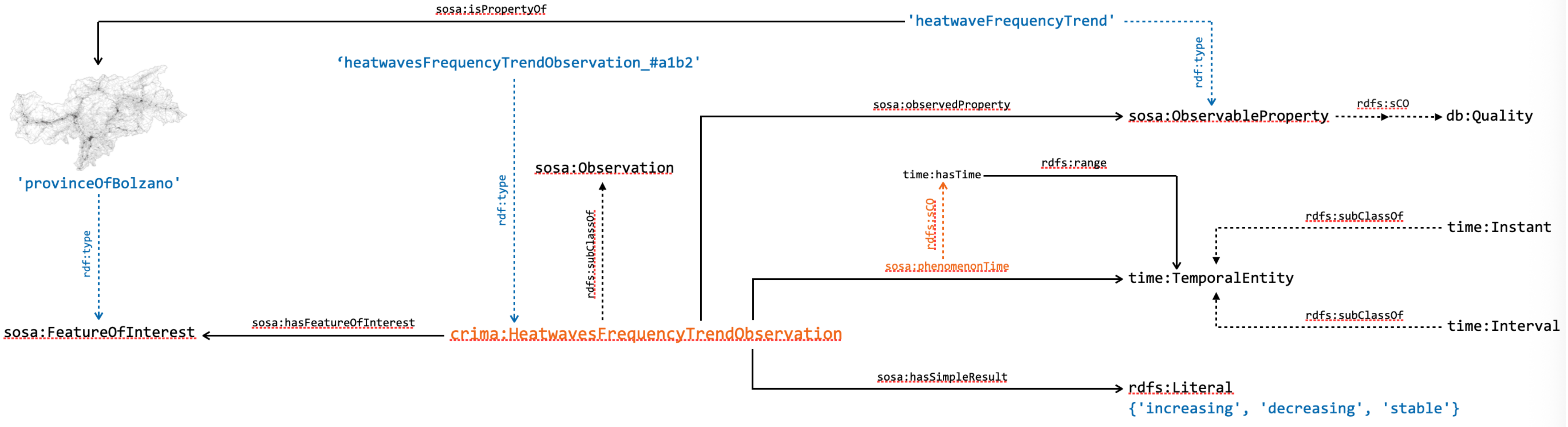
DOLCE: A descriptive ontology for linguistic and cognitive engineering[1]

Domain entities and their properties:

The SOSA 'observation' pattern

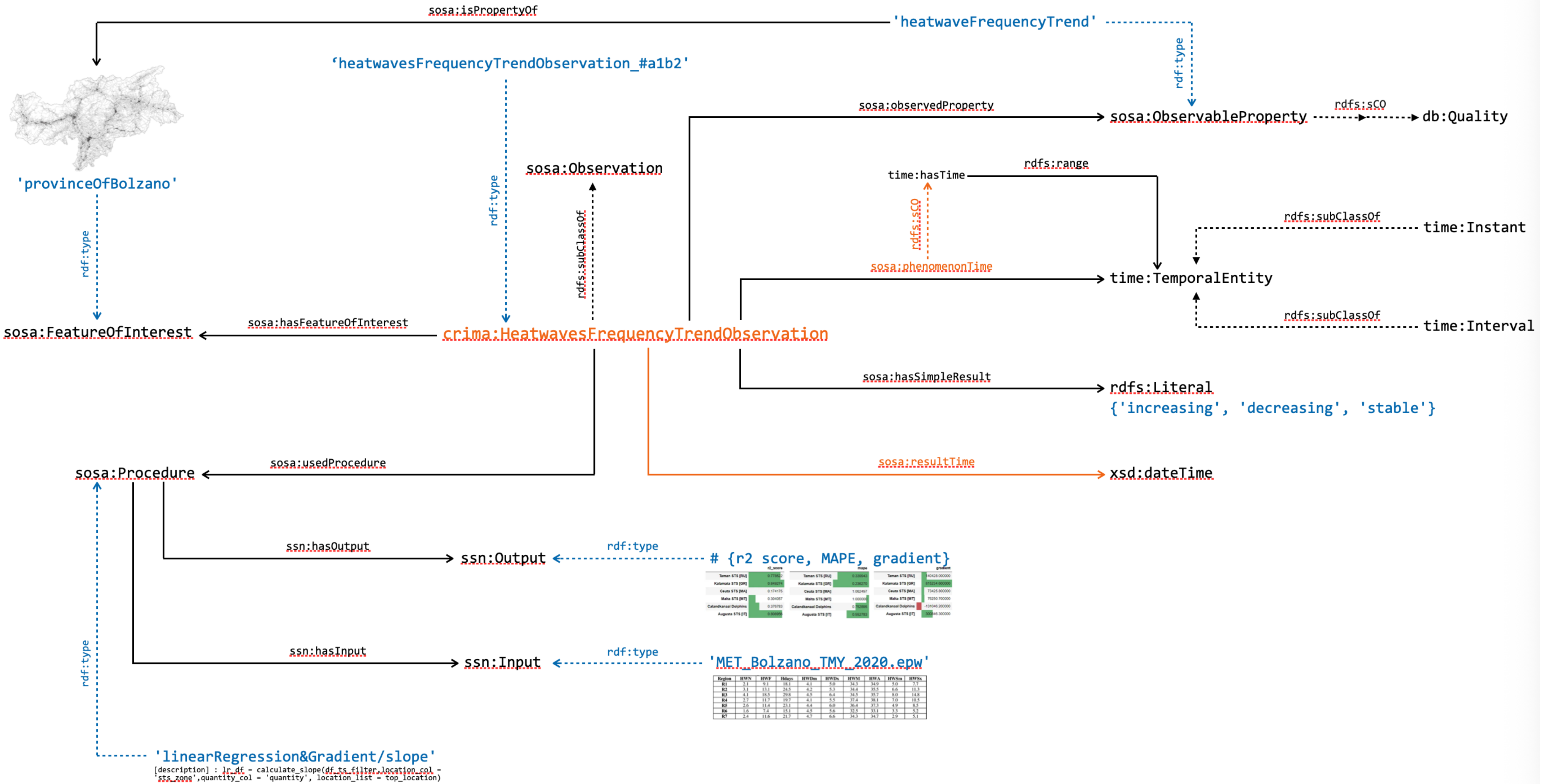
• The (increase/decrease/stationary) frequency of a given phenomenon is represented as a **property** of an entity.
• Properties' representation follow the SOSA approach (new release: March 2026): they are observed.

• **Observations** can be taken as (partial) description of states.
• Different relationships between events and **states** can be introduced: events and processes can modify the state of an entity and, vice versa, the state of an entity can represent the pre-condition for the occurrence of an event (the so-called 'initiation', 'termination' and 'allowing' relationships between states and events are mentioned later on in this presentation).



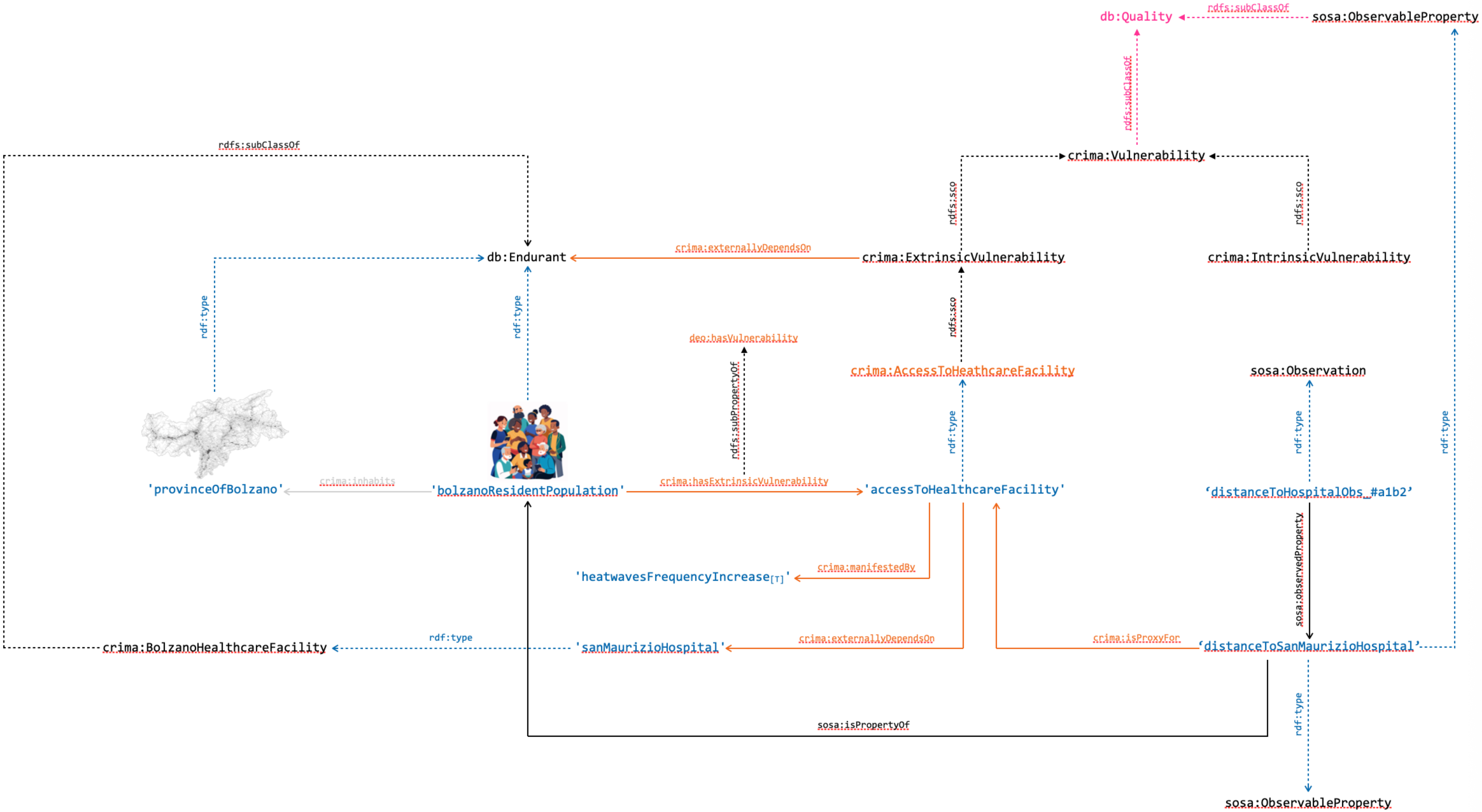
https://www.w3.org/TR/vocab-ssn/
W3C Recommendation 19 October 2017
https://w3c.github.io/sdw-sosa-ssn/ssn/
W3C Editor's Draft 05 November 2025

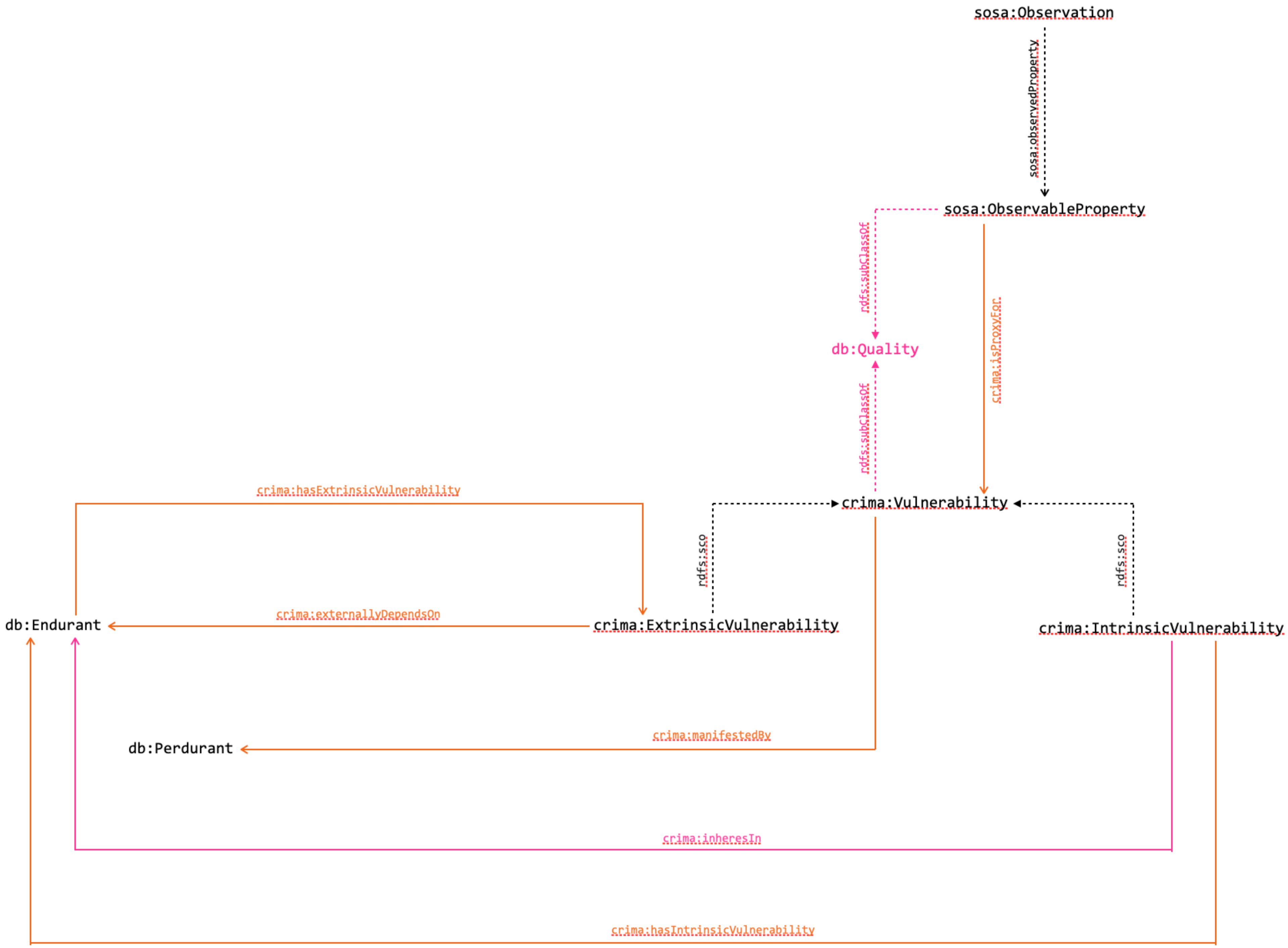
- **Procedure:** workflow, protocol, plan, algorithm, or computational method specifying how to make an Execution. A Procedure is re-usable, and might be applied in many Executions (Actuations, Observations, or Samplings). It explains the steps to be carried out to arrive at reproducible results".



Vulnerabilities as endurant properties:

Intrinsic, Extrinsic, proxies and triggering factors


db:Quality
rdfs:subClassOf
sosa:ObservableProperty
rdfs:subClassOf
rdfs:subClassOf
crima:Vulnerability
rdfs:sco
rdfs:sco
db:Endurant
crima:externallyDependsOn
crima:ExtrinsicVulnerability
crima:IntrinsicVulnerability
rdf:type
rdf:type
dep:hasVulnerability
rdfs:sco
rdfs:subPropertyOf
crima:AccessToHeathcareFacility
sosa:Observation
rdf:type
rdf:type
rdf:type
'provinceOfBolzano'
crima:inhabits
'bolzanoResidentPopulation'
crima:hasExtrinsicVulnerability
'accessToHealthcareFacility'
'distanceToHospitalObs_#a1b2'
crima:manifestedBy
'heatwavesFrequencyIncrease[T]'
sosa:observedProperty
rdf:type
crima:BolzanoHealthcareFacility
'sanMaurizioHospital'
crima:externallyDependsOn
crima:isProxyFor
'distanceToSanMaurizioHospital'
sosa:isPropertyOf
rdf:type
sosa:ObservableProperty


sosa:Observation
sosa:observedProperty
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rdfs:subClassOf
db:Quality
crima:isProxyFor
rdfs:subClassOf
crima:hasExtrinsicVulnerability
crima:Vulnerability
rdfs:sco
rdfs:sco
db:Endurant
crima:externallyDependsOn
crima:ExtrinsicVulnerability
crima:IntrinsicVulnerability
db:Perdurant
crima:manifestedBy
crima:inheresIn
crima:hasIntrinsicVulnerability

Being vulnerable and exposed as derived properties:

A reasoning example

- Assumptions:
- Entities are said to be vulnerable to hazards (or, extreme events).
- Entities are exposed to impacts (as derived from hazards).
- Impacts of hazards are meant to (negatively or positively) affect entities.
- Proxies are qualities of entities that support the definition of their vulnerabilities w.r.t. certain impacts.
- Entity vulnerabilities get manifested because of impacts (they can stay latent until certain impacts occur: assessment-wise, their score is negligible until certain contextual conditions occur).
- 'Being exposed to' is not 'being vulnerable to': the first depends on spatio-temporal conditions only, whereas the latter on spatio-temporal and specific vulnerability conditions (i.e., having at least a vulnerability for an impact in the chain of events rooted in the hazard the entity is vulnerable to).

- Example:

H(coldSpells) -leadsTo-> I(icyConditions) -leadsTo-> I(slipperyPavement)
[chain of events.root: extreme temperature decrease, frost days]

FOI(elderlyPopulation) -observableProperty-> P(physical&mentalWeakness)

P(physical&mentalWeakness) -is[positiveMonotonic]ProxyFor-> V(beingProneToFall)
['the higher the physical&mental weakness, the more prone to fall']

```
[... P(immobility) -is[positiveMonotonic]ProxyFor-> V(beingProneToFall)
     P(age) -is[positiveMonotonic]ProxyFor-> V(beingProneToFall)
     P(walkerAssisted) -isProxyFor-> V(beingProneToFall)
 ... ]
```

FOI(elderlyPopulation) -hasVulnerability-> V(beingProneToFall)
V(beingProneToFall) -manifestedBy-> I(slipperyPavement)

|= FOI(elderlyPopulation) -isVulnerableTo-> H(coldSpells)

- Notation:

H	:	hazard factor
I	:	impact factor
FOI	:	feature of interest
V	:	vulnerability factor
P	:	property
G	:	geometry
INT	:	time interval
E	:	exposure factor

• Assumptions:
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• moreover, if
FOI(elderlyPopulation) -hasGeometry-> G(x)
I(slipperyPavement) -hasGeometry-> G(y)

G(y) -covers-> G(x)
[see RCC8 support by geoSparql]

PresentAt(elderlyPopulation) -time:interval-> INT(x)
PresentAt(slipperyPavement) -time:interval-> INT(y)

INT(y) -covers-> INT(x)
[see Allen's relations supported by time:]

|= FOI(elderlyPopulation) -isExposedTo-> I(slipperyPavement)
[entity is exposed since it is present when & where the event occurs]

• by combining 'being vulnerable to' and 'being exposed to', we also have:
|= E(elderlyPopulation)
[is an exposure]

• Notation:
H : hazard factor
I : impact factor
FOI : feature of interest
V : vulnerability factor
P : property
G : geometry
INT : time interval
E : exposure factor

Impact Chain connections and their semantics

Background literature and goal

Main sources of inspiration:

- Galton (2012). States, processes and events, and the ontology of causal relations.

Additional literature:

- Toyoshima, F., Mizoguchi, R., & Ikeda, M. (2019). Causation: A functional perspective. Applied Ontology, 14(1), 43-78.
- Baratella, R., Fumagalli, M., Oliveira, Í., & Guizzardi, G. (2022). Understanding and modeling prevention. In International Conference on Research Challenges in Information Science

Goal: to provide an ontological foundation of the Impact Chain connections (Leads to, Affects, Impact on, Mitigates)

*notes in blue are considerations related to IC

**notes in black and italic are related to the reference itself*

Galton (2012). States, processes and events, and the ontology of causal relations.

Proposes an analysis of *causation* relationships between tokens, i.e. individuals, (and not *causality* - i.e. an explanatory law which holds between types) considering :(i) **states (S)**, (ii) **processes (P)**, and (iii) **events (E)**.

While some relations are properly causal (e.g. between two **events** and an **event** and a **processes**), others are rather causal-like ("enabling"), for example between **states** and **events**.

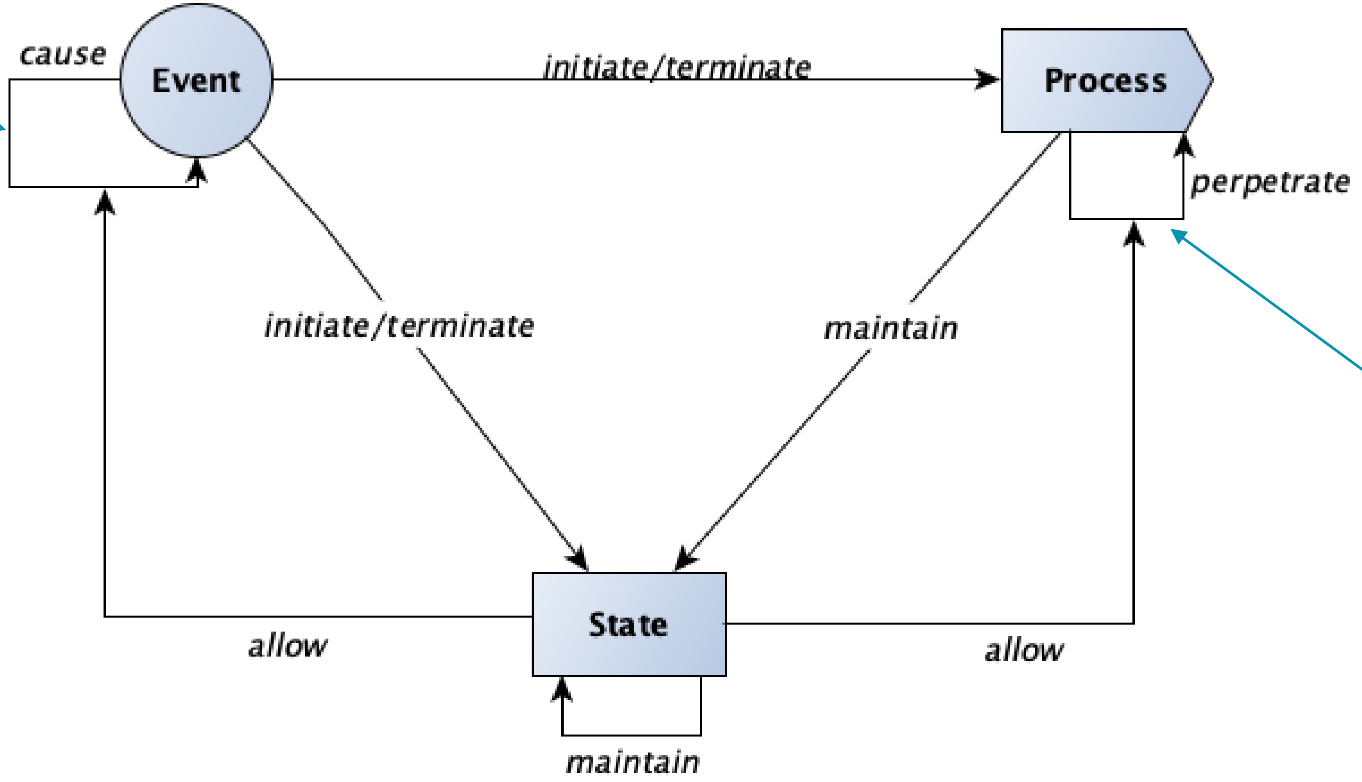
**processes* *are considered open-ended (unbounded) homogeneous entities, as a sort of dynamic states – continuant-like*

- S1 *allows* E1 (or P1)**could be used with properties; it could be applied to "Affects"*
- E1 *causes* E2*could fit with "Leads to"
- P1 *perpetuates* P2
- E1 *initiates* P1 (or S1)*this might also be condensed with "Leads to" as Hazard typically initiates an IC
- E1 *terminates* P1 (or S1)
- S1 *maintains* S2*counterpart of perpetrates

Galton (2012). States, processes and events, and the ontology of causal relations.

in the IC context we can consider "state" as sort of dynamic property

Here the **state** allows the *cause* relation to cause the **event**



Here the **state** allows the *perpetrate* relation to perpetrate the **process**

Readapted from Galton 2012, Fig 6

allows* could be applied to types (classes) and could also be negativethis could be applied to IC connection "Mitigates"*

Toyoshima et al. (2019). Causation: A functional perspective

Advances a **functional**, **context**-, -**goal** dependent *theory of causation* considering as relata **states (S)**, **events (E)**, and **processes (P)** (following Galton 2012), which more strongly associated *change* with causation (if there is a dependence between the two).**Perhaps the theory if more suitable for an artefact-centric domain, although the theory should be generalisable*

- *achieves* ("cause achieves the effects" [p. 13], achieving a ***goal*** because of a ***function***)*this might be somewhat similar to "Leads to," yet I am not convinced about the label as it carries intentional assumptions
- *prevents**counterpart of *achieves*. Could be useful to explore for "Mitigates"
- *allows**useful to capture the mediation of states (for me properties, such as vulnerability) in causal-like relationships
- *disallows*

achieves (and *prevents*) are proper causal relations and define causally efficacious occurrents (i.e., event, state, process)

allows (and *disallows*) support properly causal relations as **causally-mediated/precondition-based** relations, i.e., indirect causal relations

here* ***processes* *are considered in a more classic ontological way, i.e. occurrent/perdurant (extended in time)*
goal *is "a state to be achieved" [p. 9] and can be intentional or not*

Toyoshima et al. (2019). Causation: A functional perspective

Causal relation **(achieves/prevents)**: central notions on ***achievement*** (goal-oriented, intentional and unintentional), "how to achieve," a.k.a. the ***function***, and "what to achieve," a.k.a. the state that play the role of a ***goal*** within the ***context*** of a ***system*** (the achievement is always dependent on the context).
Example: heart pumping blood as a device (pressure, transferring blood, circulation, beating)
*n*ote that achieves* (thus *prevents*) applies to: (E, S), (P, P), (P, S), (S, S)

State-mediated causation: the theory assumes a strong role of states in causal relations and change, as change is a "transition between states" [p. 18]. States can be cause and affects (see e.g. *achieves*) and no change can happen without the change of state.

States are seen as *preconditions/facilitators* for a causal relation (á *la* Galton), thus having an *indirect* role in the ***causal chain***, and similarly to *achieve*/*prevent* configuration, states can also "prevent" when it is ***incompatible*** with another state in respect to a context.

**in the IC context we can consider "state" as a sort of dynamic property - yet in this case I do not know about their proper causal role, maybe an approach à la Galton (2012) might be more suitable*

Toyoshima et al. (2019). Causation: A functional perspective

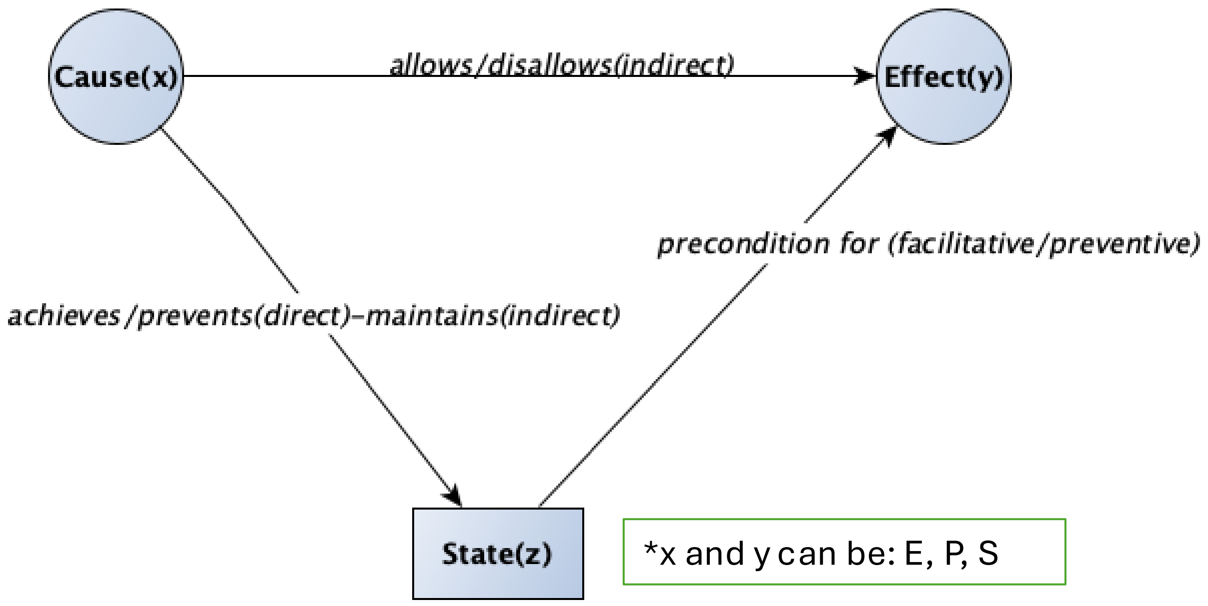
**in the IC context we can consider "state" as a sort of dynamic property - yet in this case I do not know about their proper causal role, maybe an approach à la Galton (2012) might be more suitable*

Functional causal relations (p. 22)

	Positive	Negative
Direct	*achieves*	*prevents*
Indirect	*allows*	*disallows*

Readapted from Toyoshima et al. 2019, Table 1

Configuration of State-mediated causation (p.20)



Readapted from Toyoshima et al. 2019, Fig 1

Baratella, R. et al. (2022)Understanding and modeling prevention

Offers an ontological model of **prevention** (i.e. blocking event while is happening/interrupting its unfolding while its happening). This work is *Unified Foundational Ontology*-based and rely on the notion of **dispositions** and their relations with **events** seen as manifestation of those object's dispositions, such as fragility, vulnerability, capacity.

Key aspects of prevention:

- Dispositions interacts with each other (**Mutual Activation Pattern - MAP**)
- To prevent an event the options are:
  - to **modify the disposition** that the object bears
  - to **remove the object** from the context (*situation* in UFO)
  - to **remove the disposition** that is *paired* in a MAP
  - to introduce an **incompatible disposition**

Prevention can be direct and indirect: (I) **antidote** and (ii) **countermeasure**

*useful for "Mitigates". Although we do not have *situation* and *disposition*, this characterisation is useful to define that properties attached to objects and the participation of exposures in causal mechanisms (*situation activates disposition manifested in event*). Also note that in this representation are objects (endurants) that creates and terminates events.

Implications for the IC connections

- ***Leads to***

"Hazards function as the initiating factor in the impact chain. They can trigger both hazards and direct impacts. For instance, torrential rain may trigger landslides, which can damage infrastructure."

Leads to is the only "genuine" causal relation between types of perdurants that connects as relata:

- Hazard, Hazard
- Hazard, Impact
- Impact, Impact

- **Affects**

"Lack of land ownership (vulnerability) affects the disruption of agricultural livelihoods in a flood (impact)."

Affects is a sort of property mediated connection (PMC), following the characterisation of Toyoshima et al. and inspired by Galton.
Affects is not a proper causal relation between types, but rather a causal-like relation in which 1 or more conditions set the stage for an Impact or Risk. The relata of *Allows* are indeed:

- Vulnerability, Impact/Risk

Implications for the IC connections

-Impacts on
“An increase in power outages (impact) impacts industries (exposure).”

Impacts on is a **goal-based causal relation** between types, in this case a perdurant and an endurant. Differently from *Leads to* and *Affects* this relation is explicitly grounded on the goals of the assessor(s) as the very notion of “impact” is dependent upon the values and goals of an agent. This representation can be derived by the several UFO-based analyses (e.g. COVER). The relata of Impacts on are:

- Impact, Exposure

Beside the goal-based characterisation of *Impacts on*, it explicitly include the participants of a certain perdurant, i.e. the Exposure, implying that while a simple causal relation connects perdurants, an impact can be only directed to objects.

Implications for the IC connections

-***Mitigates***
"An improved irrigation technique (adaptation measure) mitigates the decrease in crop yield (impact) during droughts.”

Mitigates is unique as it is the only connection that has a “negative” but direct effect (in the sense of countermeasure and prevention); it is also another **goal-based** relation as it assumes that there are states that an assessor desire to reduce. This connection relates:

- Adaptation/Risk Reduction Measure, Vulnerability
- Adaptation/Risk Reduction Measure, Impact
- Adaptation/Risk Reduction Measure, Risk

*Is "Adaptation" or "Risk Reduction Measure"?

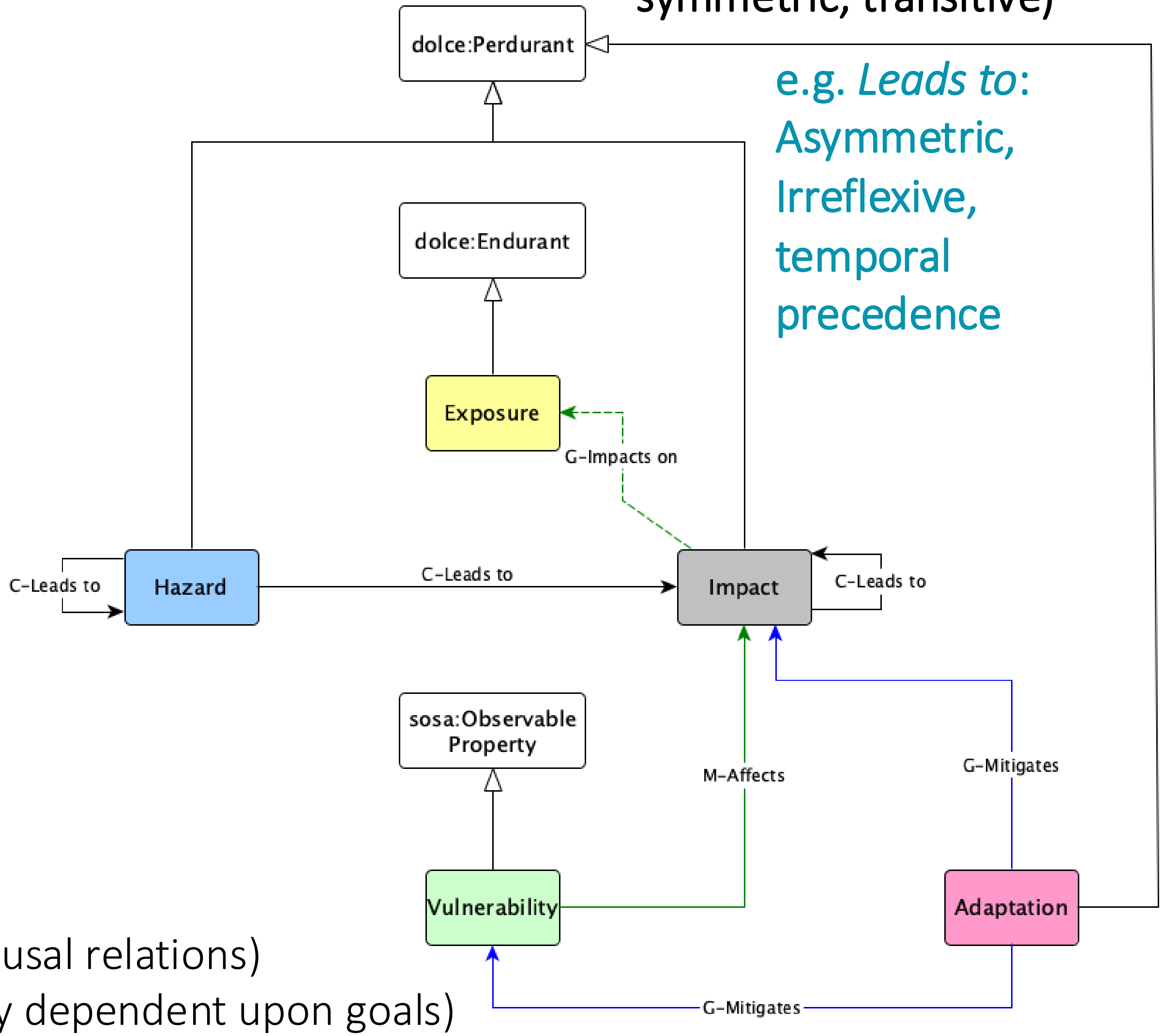
Implications for the IC connections

IC connection properties? (e.g. functional, inverse, symmetric, transitive)

	Positive	Negative
Direct	*Leads to (neutral?), Impacts on (goal-based)*	*Mitigates (goal-based)*
Indirect	*Affects (goal-based)*	

Table inspired by Toyoshima et al. 2019

*the interactions between properties (e.g. of the hazards and the exposures) influence the chain



e.g. *Leads to*: Asymmetric, Irreflexive, temporal precedence

C: Proper causal relation (between perdurants)
M: Mediated relation (the property allows the causal relations)
G: Goal-based relation (the relations are explicitly dependent upon goals)

*Risk factor?

CRIMA

Impact classification:

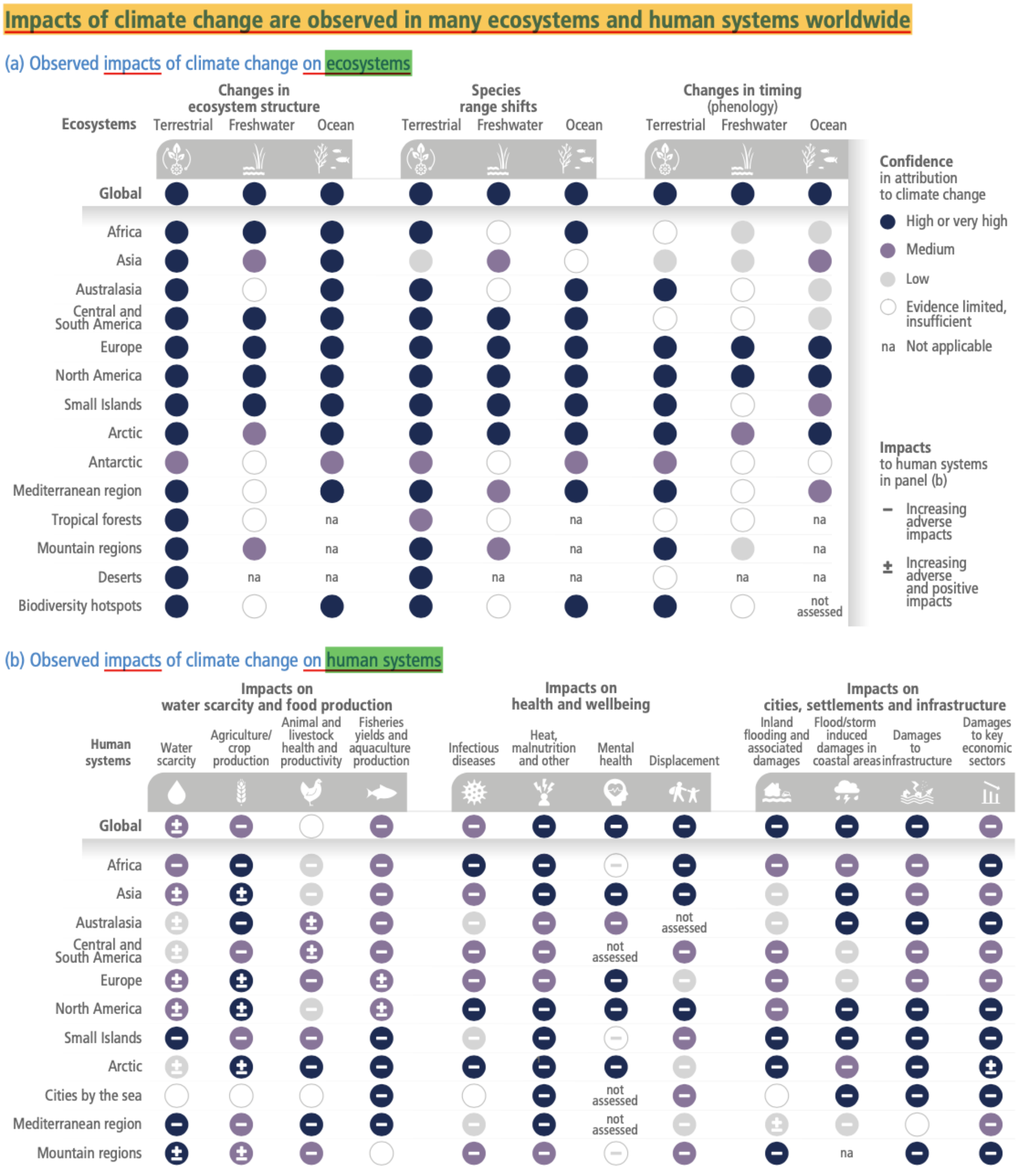
The IPCC WGII AR6 proposal

Climate Change 2022: Impacts, Adaptation, and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press, Cambridge, UK and New York, NY, USA, pp. 3-33, doi:10.1017/9781009325844.001.

Climate Change 2022:

Impacts, Adaptation and Vulnerability

Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change



Observed global and regional impacts on ecosystems and human systems attributed to climate change. Confidence levels reflect uncertainty in attribution of the observed impact to climate change. Global assessments focus on large studies, multi-species, meta-analyses and large reviews. For that reason they can be assessed with higher confidence than regional studies, which may often rely on smaller studies that have more limited data. Regional assessments consider evidence on impacts across an entire region and do not focus on any country in particular.

(a) Climate change has already altered terrestrial, freshwater and ocean **ecosystems** at global scale, with multiple impacts evident at regional and local scales where there is sufficient literature to make an assessment. Impacts are evident on ecosystem structure, species geographic ranges and timing of seasonal life cycles (phenology).

(b) Climate change has already had diverse adverse impacts on **human systems**, including on water security and food production, health and well-being, and cities, settlements and infrastructure.